

✓ Assignment-9

Aim- Building a Simple Data Science Pipeline: Integrate all learned techniques in previous labs into a complete workflow.

Theory

A Data Science pipeline is a structured process used to convert raw data into meaningful insights. It typically involves steps such as data ingestion, cleaning, transformation, analysis, and modeling. This assignment demonstrates a basic pipeline using the Iris dataset.

The pipeline begins with **Data Ingestion**, where the Iris dataset is loaded. This is followed by **Data Cleaning** (checking for missing values). Then, **Data Wrangling** involves creating a new feature (`petal_area`). **Exploratory Data Analysis (EDA)** is performed using visualizations (scatter plots, histograms) and correlation analysis.

Next, **Feature Scaling** is applied using `StandardScaler`. Finally, a **Linear Regression Model** is built and evaluated using metrics like Mean Squared Error (MSE) and R^2 score, demonstrating a complete workflow from raw data to a predictive model.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)

print("Data Loaded:")
print(df.head())

print("\nMissing Values:\n", df.isnull().sum())

df['petal_area'] = df['petal length (cm)'] * df['petal width (cm)']

plt.figure(figsize=(5,4))
plt.scatter(df['sepal length (cm)'], df['petal length (cm)'])
plt.title("Sepal Length vs Petal Length")
plt.xlabel("Sepal Length")
plt.ylabel("Petal Length")
plt.grid(True)
plt.show()

plt.figure(figsize=(5,4))
plt.hist(df['petal length (cm)'], bins=10)
plt.title("Petal Length Distribution")
plt.show()

corr = np.corrcoef(df['sepal length (cm)'], df['petal length (cm)'])[0,1]
print("\nKarl Pearson Correlation Coefficient:", round(corr,4))

X = df[['sepal length (cm)']]
y = df['petal length (cm)']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

model = LinearRegression()
model.fit(X_scaled, y)

y_pred = model.predict(X_scaled)

mse = mean_squared_error(y, y_pred)
r2 = r2_score(y, y_pred)

print("\nRegression Equation: y =", round(model.coef_[0],2), "x +", round(model.intercept_,2))
print("Mean Squared Error (MSE):", round(mse,4))
print("R² Score:", round(r2,4))

plt.scatter(X_scaled, y, label="Actual Data")
plt.plot(X_scaled, y_pred, color='red', label="Regression Line")
plt.xlabel("Scaled Sepal Length")
plt.ylabel("Petal Length")
plt.title("Linear Regression Fit")
plt.legend()
plt.grid(True)
```

```
plt.show()
```

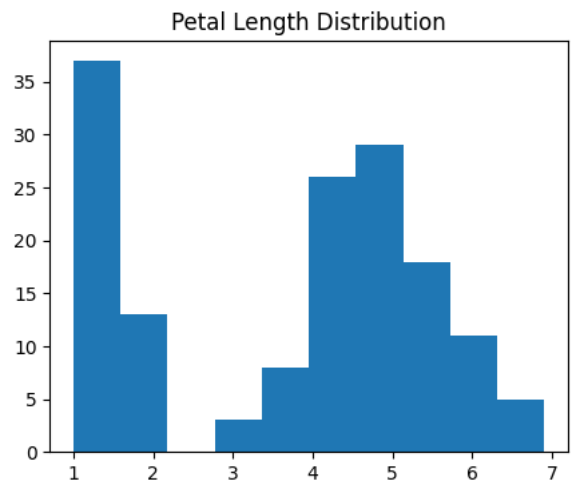
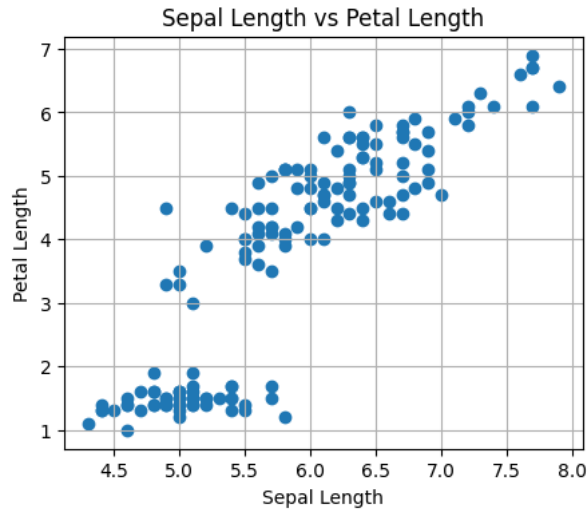
Data Loaded:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2

Missing Values:

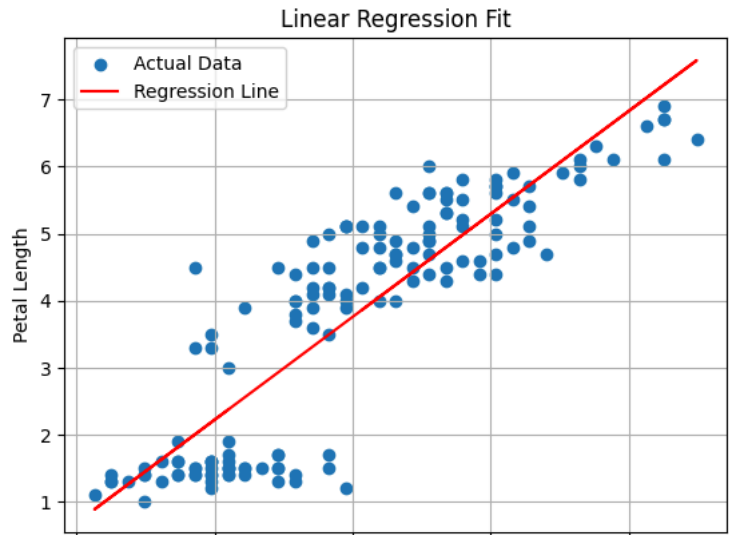
sepal length (cm)	0
sepal width (cm)	0
petal length (cm)	0
petal width (cm)	0

dtype: int64



Karl Pearson Correlation Coefficient: 0.8718

Regression Equation: $y = 1.53x + 3.76$
Mean Squared Error (MSE): 0.7431
 R^2 Score: 0.76





The Linear Regression model achieved an R^2 Score of 0.76, indicating that approximately 76% of the variance in petal length can be explained by the sepal length. The Mean Squared Error (MSE) of 0.7431 suggests the average squared difference between the actual and predicted petal lengths is relatively low, indicating a reasonably good fit for this simple linear model.

1. What is a Data Science pipeline? Explain its key stages with examples.

A Data Science pipeline is a structured sequence of steps used to collect, process, analyze, and model data to extract meaningful insights or make predictions. It ensures that raw data is transformed into useful information in a systematic and efficient manner.

Based on your previous practicals (data ingestion, cleaning, wrangling, visualization, and modeling), a typical pipeline includes:

Data Ingestion: Collecting data from sources such as CSV files, APIs, or web scraping. For example, loading the Titanic dataset using `pandas.read_csv()` as done in Assignment 2 . Data Cleaning: Handling missing values, duplicates, and inconsistencies using functions like `isnull()`, `dropna()`, and `fillna()` as performed in your data cleaning practical . Data Wrangling / Transformation: Structuring and transforming data, such as creating new columns like "Status" based on scores in Assignment 3 . Exploratory Data Analysis (EDA): Understanding data using statistics and visualizations like scatter plots, histograms, and box plots using Matplotlib and Seaborn . Feature Engineering: Creating meaningful features to improve model performance. Model Building: Applying algorithms such as linear regression (Assignment 8). Evaluation: Measuring model performance using metrics like MSE and R^2 .

Thus, a data science pipeline connects all steps into a complete workflow from raw data to final insights.

2. What is Exploratory Data Analysis (EDA)? How do univariate and bivariate analyses help in understanding the dataset?

Exploratory Data Analysis (EDA) is the process of analyzing and summarizing a dataset to understand its main characteristics before applying machine learning models. It helps in identifying patterns, detecting outliers, and checking data quality.

According to your practical, EDA uses visualization and statistical techniques to simplify complex data and support decision-making .

Univariate Analysis: It analyzes a single variable at a time. It helps understand distribution, central tendency, and spread of data.

Example: Histogram of petal length in the iris dataset (Assignment 5) . Bivariate Analysis: It analyzes the relationship between two variables. It helps identify correlation or patterns between variables. Example: Scatter plot of sepal length vs sepal width to observe relationships .

EDA is important because it helps in making informed decisions before building models.

3. Differentiate between Label Encoding and One-Hot Encoding. When should each be used?

Label Encoding and One-Hot Encoding are techniques used to convert categorical data into numerical form so that machine learning models can process it.

Label Encoding: Assigns a unique integer to each category. Example: Male = 0, Female = 1. Use: When categories have an inherent order (e.g., Low, Medium, High). One-Hot Encoding: Converts categories into multiple binary columns (0 or 1). Example: Male $\rightarrow$ [1, 0], Female $\rightarrow$ [0, 1]. Use: When categories are nominal (no order), to avoid misleading relationships.

Thus, label encoding is simple but may introduce false relationships, while one-hot encoding preserves independence between categories.

4. Why is feature scaling required? Compare normalization and standardization.

Feature scaling is the process of adjusting the range of data values so that all features contribute equally to the model. It is important because many machine learning algorithms are sensitive to the scale of data.

Normalization (Min-Max Scaling): Scales data to a fixed range, usually 0 to 1.

Formula:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Use: When data does not follow a normal distribution.

Standardization (Z-score Scaling): Transforms data to have mean = 0 and standard deviation = 1.

Formula:

$$X' = \frac{X - \mu}{\sigma}$$

Use: When data follows a normal distribution.

Normalization is useful for bounded data, while standardization is preferred for most machine learning models.

5. What is feature engineering? Explain how creating new features can improve model performance.

Feature engineering is the process of creating new input features or modifying existing ones to improve the performance of machine learning models.

In your data wrangling practical, a new column "Status" was created based on score values, which is an example of feature engineering .

Creating new features improves model performance because:

It provides more meaningful information to the model It helps capture hidden patterns in data It improves prediction accuracy It reduces noise and redundancy

Example: Instead of using "Date" directly, extracting "Month" or "Day" can help the model understand seasonal patterns better.

Conclusion

In this assignment, a complete Data Science pipeline was demonstrated using the Iris dataset. We performed data ingestion, cleaning, feature engineering, exploratory data analysis, feature scaling, and built a linear regression model. This structured approach highlights how raw data can be processed and transformed into a predictive model, integrating various data science techniques into a coherent workflow.